

# **“Preventing Wrong Decisions in Smart Building Systems”**

## ***Minor Project Report***

*submitted in partial fulfillment of the  
requirements for the award of the degree of*

**BACHELOR OF TECHNOLOGY**

in

**COMPUTER SCIENCE & TECHNOLOGY**

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**May – 2026**

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We hereby certify that the project work entitled “**Preventing Wrong Decisions in Smart Building Systems**” in partial fulfilment of the requirements for the award of the Degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING with specialization in Cybersecurity and Digital Forensic and submitted to the Department of Systemics, School of Computer Science, University of Petroleum & Energy Studies, Dehradun, is an authentic record of my our work carried out during a period from **Jan, 2026** to **May, 2026** under the supervision of **Amit Gurung, \_\_\_\_Cluster, UPES**.

The matter presented in this project has not been submitted by us for the award of any other degree of this or any other University.

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Guide

Project

# Acknowledgement

We sincerely thank our respected Head Department, **Mr. Hitesh Kumar** for his great support in doing our project.

We sincerely express our gratitude to **Dr. Srinivasan Ramachandran**, Faculty, School of Computer Science (AI and Computer Vision Cluster), UPES, for his constant guidance and support. We especially thank him for providing the necessary hardware resources and for his valuable mentorship in helping us understand and implement IoT technologies throughout this project.

We are also grateful to Dean SoCS UPES for giving us the necessary facilities to carry out our project work successfully. We also thank our Course Coordinator, **Azra Nazir** and our Activity Coordinator **Nitesh Kumar Singh** for providing timely support and information during the completion of this project.

We would like to thank all our friends for their help and constructive criticism during our project work. Finally, we have no words to express our sincere gratitude to our parents who have shown us this world and for every support they have given us.

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# Abstract

The rapid growth of smart building systems and IoT-based automation has increased reliance on sensor-driven decision-making processes. Most existing systems focus on securing data transmission using protocols such as MQTT with TLS, ensuring that data is protected during communication. However, these systems assume that incoming sensor data is always correct. In reality, sensors can be faulty or manipulated while still sending believable values, leading to incorrect decisions without any warning. This can result in energy inefficiency, system malfunction, and potential safety risks.

To address this issue, the proposed Hybrid IoT Intrusion Detection System (IDS) introduces a mechanism to verify both the security and correctness of data. The system collects real-time data from multiple IoT sensor nodes connected through ESP32 devices and an MQTT broker. A real-world attack scenario was also performed using ARP spoofing, allowing live interception and manipulation of sensor data. This helped in generating realistic datasets for training and testing the system.

The system combines multiple detection approaches, including rule-based logic and machine learning, to identify abnormal patterns in sensor data over time. Instead of relying on single readings, it analyses sequences of data to detect subtle changes such as sudden spikes, gradual drift, repeated patterns, or unexpected drops. In addition, a network monitoring layer observes traffic behaviour to detect suspicious activities before they affect the system.

All these components work together in a hybrid decision process, improving the accuracy and reliability of anomaly detection. The system also includes real-time visualization to help monitor sensor behaviour and identify issues easily.

Overall, the proposed solution ensures that automated decisions are based on trustworthy data rather than blindly accepted inputs. It provides a practical and scalable approach to improving the reliability and security of smart building systems and other IoT-based environments.

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# Chapter 1

## Introduction

### 1.1 History

The evolution of building management systems has progressed from manual control mechanisms to fully automated smart environments powered by IoT technologies. Earlier systems relied on human intervention to monitor and regulate environmental conditions such as temperature and humidity. While these systems were simple, they lacked efficiency and scalability.

With the advancement of IoT, smart building systems began using distributed sensors and automated controllers to make real-time decisions. Technologies such as ESP32-based sensor nodes and MQTT communication enabled continuous data collection and remote monitoring. These systems improved operational efficiency and reduced human effort.

However, as automation increased, so did the dependency on sensor data. Most modern systems focus primarily on securing communication channels to ensure data Authenticity during Transmission. Despite this, they assume that the received data is always correct. In reality, sensors can malfunction or be manipulated, resulting in incorrect but believable data being used for decision-making.

This limitation led to the need for systems that not only secure data transmission but also validate the correctness of data. Recent advancements in cybersecurity and machine learning have enabled the development of intelligent detection systems capable of identifying anomalies and malicious behavior in real time.

The proposed system builds upon these developments by combining IoT-based data collection with hybrid intrusion detection techniques, ensuring that automated decisions are based on reliable and trustworthy data.

### 1.2 Requirement Analysis

The proposed system requires a combination of IoT infrastructure, network monitoring, and intelligent data analysis to detect anomalies and potential attacks in real time. The requirements focus on ensuring accurate data collection, reliable detection mechanisms, and secure system operation. These requirements are categorized into functional and non-functional requirements.

#### Functional Requirements

The system must collect real-time environmental data from multiple IoT sensors (ESP32 with DHT11).

- The system must transmit sensor data to a central server using MQTT protocol.
- The system must support real-time monitoring of incoming sensor data.
- The system must detect abnormal patterns in data using rule-based logic (spike, drift, drop, noise).
- The system must implement machine learning to classify normal and attack conditions.

- The system must analyze data in temporal sequences rather than individual readings.
- The system must detect replay attacks by comparing current data patterns with historical sequences.
- The system must monitor network traffic using an intrusion detection system (Snort).
- The system must generate alerts for suspicious activities at both network and data levels.
- The system must integrate multiple detection methods into a unified decision system.
- The system must provide visualization of sensor data and detected anomalies.

## **Non-Functional Requirements**

- **Security:** The system must ensure secure communication and detect malicious data manipulation.
- **Reliability:** The system must provide consistent and accurate detection under continuous operation.
- **Scalability:** The system should support additional sensors and nodes in the future.
- **Performance:** The system must process and analyze data in real time with minimal delay.
- **Accuracy:** The system should minimize false positives and false negatives in detection.
- **Usability:** The system should provide clear and understandable output through visualization.
- **Robustness:** The system should handle noisy, incomplete, or inconsistent data effectively.

### **1.3 Main Objective**

The main objective of this project is to design and develop a hybrid intrusion detection system for IoT-based smart building environments that ensures reliable and secure decision-making. The system aims to detect anomalies and malicious activities in real time by analyzing both sensor data and network behavior, preventing incorrect decisions caused by faulty or manipulated data.

### **1.4 Sub Objectives**

- To collect real-time environmental data using IoT devices such as ESP32 and DHT11 sensors.
- To establish reliable communication between devices using MQTT protocol.
- To simulate real-world attacks such as ARP spoofing for generating realistic datasets.
- To detect abnormal patterns in sensor data using rule-based techniques.
- To develop a machine learning model for classifying normal and attack conditions.
- To analyze temporal behavior of data using sequence-based processing.
- To implement replay attack detection using pattern comparison techniques.
- To monitor network traffic using an intrusion detection system (Snort).
- To integrate network-level and data-level detection into a unified system.
- To provide real-time visualization of system behavior and detected anomalies.

### **1.5 System Workflow**

The system follows a structured workflow to collect, analyze, and detect anomalies in IoT-based smart building environments. The process integrates data collection, attack simulation, detection mechanisms, and final decision-making.

- **Data Collection:** Environmental data such as temperature and humidity is collected from multiple DHT11 sensors connected to ESP32 devices.
- **Data Transmission:** The collected data is transmitted in real time to an MQTT broker for centralized processing.
- **Attack Simulation:** A real-world attack scenario is created using ARP spoofing to intercept and manipulate sensor data during transmission.
- **Data Processing:** The incoming data is cleaned and prepared for analysis, including grouping into sequences for temporal evaluation.
- **Feature Extraction:** Important statistical and temporal features are extracted from the data to identify patterns and behavior.
- **Detection (Rule-Based):** Basic anomalies such as spikes, drift, noise, and drops are detected using predefined rules.
- **Detection (Machine Learning):** A trained Random Forest model classifies the data into normal or different attack categories.
- **Replay Detection:** The system compares current data patterns with past sequences to detect repeated or replayed data.
- **Network Monitoring:** Snort IDS monitors network traffic to identify suspicious activities such as scanning or unauthorized access.
- **Decision Making:** All detection outputs are combined in a hybrid decision system to generate a final result.
- **Output and Visualization:** The system displays results through real-time graphs and alerts for detected anomalies.

## **`Chapter 2**

# **SYSTEM ANALYSIS**

### **2.1 Introduction**

The increasing adoption of IoT-based systems in smart buildings has enabled automated monitoring and control of environmental conditions using sensor data. These systems rely heavily on continuous data streams from multiple devices to make real-time decisions, improving efficiency and reducing manual effort.

However, most existing systems focus mainly on securing communication channels while assuming that the received data is always correct. In practice, sensor data can be faulty, noisy, or even intentionally manipulated through cyber-attacks. This leads to situations where systems function normally but operate on incorrect information, resulting in poor decisions and potential risks.

To address these challenges, there is a growing need for intelligent systems that can detect anomalies not only at the data level but also at the network level. By combining IoT data analysis with cybersecurity techniques, it becomes possible to identify both abnormal sensor behavior and suspicious network activities.

This chapter analyzes the current approaches used in IoT systems, highlights their limitations, and explains the need for a hybrid detection system that improves reliability and security in smart building environments.

### **2.2 Existing System**

The rapid growth of Internet of Things (IoT) systems has introduced significant security challenges due to continuous communication between connected devices. Intrusion Detection Systems (IDS) have been widely used to monitor network activities and detect malicious behavior in IoT environments. IDS plays a crucial role in identifying abnormal patterns and protecting systems from cyber threats [1].

Existing IDS approaches are mainly classified into signature-based and anomaly-based detection systems. Signature-based systems detect known attacks using predefined patterns, while anomaly-based systems identify deviations from normal behavior, making them capable of detecting unknown threats but often producing higher false positives. Suganthi et al. discussed the challenges of intrusion detection in IoT due to device heterogeneity and network complexity [2].

Recent research has focused on integrating machine learning techniques into IDS to improve detection accuracy. Rahman et al. highlighted that machine learning models can effectively detect complex attack patterns in IoT networks by learning from historical data [3].

Furthermore, Kikissagbe et al. demonstrated that machine learning-based intrusion detection systems enhance detection performance by adapting to evolving threats and reducing false alarms [4].

Hybrid intrusion detection approaches have also been proposed to combine multiple detection techniques. Alakhras et al. emphasized that hybrid IDS models improve scalability and detection accuracy by integrating different strategies such as rule-based and machine learning methods [5].

Despite these advancements, most existing systems focus either on network-level monitoring or data-level anomaly detection independently. This separation limits their effectiveness in real-world IoT environments where both network behavior and sensor data must be analyzed together. Additionally, challenges such as real-time processing, scalability, and handling dynamic IoT environments remain open research problems [6].

These limitations highlight the need for a hybrid and integrated approach that combines multiple detection techniques to improve the reliability and security of IoT-based systems.

### 2.3 Research Gap

The analysis of existing systems reveals that most intrusion detection approaches focus either on network-level monitoring or data-level anomaly detection independently. Signature-based systems are limited to detecting known attacks, while anomaly-based and machine learning approaches require large datasets and often struggle with real-time implementation.

Additionally, existing solutions lack integration between sensor data analysis and network traffic monitoring, which is critical in IoT environments where both data manipulation and network-based attacks can occur simultaneously.

Furthermore, limited research addresses real-time attack scenarios such as ARP spoofing in practical IoT deployments. Many systems are tested on simulated datasets rather than real-world manipulated data.

To overcome these limitations, there is a need for a hybrid system that combines rule-based detection, machine learning, and network-level monitoring into a unified framework. The proposed system addresses this gap by integrating IoT data analysis with network intrusion detection and real-time attack simulation, improving both accuracy and reliability.

### 2.4 Comparative Analysis

| Citation                  | Detection Type  | ML Usage    | Network Monitoring | Remarks                                     |
|---------------------------|-----------------|-------------|--------------------|---------------------------------------------|
| [1] Sarhan et al.         | Anomaly-Based   | Applied     | Not Applied        | Focus on feature selection for ML-based IDS |
| [2] Hindy et al.          | ML-Based        | Applied     | Not Applied        | MQTT-based dataset and attack detection     |
| [3] Vitorino et al.       | ML Comparison   | Applied     | Not Applied        | Comparative study of ML techniques          |
| [4] Ali et al.            | Hybrid ML       | Applied     | Applied            | Multi-layer detection for ARP spoofing      |
| [5] Roesch et al. (Snort) | Signature-Based | Not Applied | Applied            | Strong network traffic monitoring           |
| [6] Traditional IDS       | Signature-Based | Not Applied | Applied            | Detects known attacks only                  |

# Chapter 3

## System Design

### 3.1 Proposed System

The proposed system is a Hybrid Intrusion Detection System (IDS) designed for IoT-based smart building environments to ensure secure and reliable decision-making. The system integrates IoT data collection, network monitoring, and intelligent anomaly detection into a unified framework.

The system begins with multiple IoT sensor nodes (ESP32 with DHT11 sensors) deployed in different zones to collect environmental data such as temperature and humidity. These sensors continuously transmit real-time data to an MQTT broker, enabling centralized monitoring and processing.

To simulate real-world attack scenarios, a Man-in-the-Middle (MITM) attack is performed using ARP spoofing, allowing interception and manipulation of live sensor data. This approach enables the generation of realistic datasets, improving the effectiveness of the detection system.

The proposed system incorporates a multi-layer detection mechanism:

- A rule-based detection layer identifies simple anomalies such as sudden spikes, gradual drift, noise, and abnormal drops in sensor data.
- A machine learning layer uses a Random Forest model trained on real-time generated datasets to classify normal and attack conditions. Machine learning-based IDS have been widely used in IoT environments to improve detection accuracy and adaptability.
- A replay detection mechanism compares current data sequences with historical patterns to detect repeated or replayed data injections.
- A network monitoring layer using Snort IDS analyzes network traffic to detect suspicious activities such as scanning, probing, and unauthorized access. IDS systems play a crucial role in identifying abnormal behavior in IoT networks.

All detection components are integrated into a hybrid decision system, which combines outputs from different layers to generate a final decision with higher accuracy and reduced false positives. Research shows that hybrid IDS approaches improve detection performance by combining multiple techniques.

The system also includes a real-time visualization module that provides graphical representation of sensor data and detected anomalies, enabling easier monitoring and analysis.

Overall, the proposed system provides a scalable and efficient solution for detecting both data-level anomalies and network-level attacks, ensuring that automated decisions are based on reliable and trustworthy information in IoT environments.

### 3.2 System Architecture

The proposed system follows a modular and layered architecture that integrates IoT data collection, network monitoring, and intelligent anomaly detection. Each component works together to ensure accurate detection of both data-level and network-level attacks in real time. The system consists of the following main components:

- **IoT Layer:**

This layer includes ESP32 devices connected with DHT11 sensors that continuously collect environmental data such as temperature and humidity from different zones.

- **Communication Layer (MQTT Broker):**

Sensor data is transmitted in real time to an MQTT broker, which acts as a central communication hub between IoT devices and the processing system.

- **Attack Simulation Layer:**

This layer performs ARP spoofing-based Man-in-the-Middle (MITM) attacks to intercept and manipulate sensor data during transmission, enabling realistic dataset generation.

- **Data Processing Layer:**

The collected data is cleaned and organized into sequences for further analysis. Feature extraction is performed to identify meaningful patterns in the data.

- **Detection Layer:**

This layer consists of multiple detection mechanisms:

1. Rule-based detection for identifying simple anomalies
2. Machine learning model (Random Forest) for classification
3. Replay detection for identifying repeated data patterns

- **Network Monitoring Layer:**

A Snort-based IDS monitors network traffic and detects suspicious activities such as scanning, probing, and unauthorized access.

- **Decision Layer:**

All detection outputs are combined in a hybrid decision system to produce a final result with improved accuracy and reduced false positives.

- **Visualization Layer:**

The system provides real-time graphs and outputs to monitor sensor behavior and detect anomalies.

### 3.3 Modules

The system is divided into multiple modules, each responsible for a specific task to ensure efficient detection of anomalies and attacks.

- **IoT Data Collection Module:**

This module collects real-time environmental data from ESP32 devices connected with DHT11 sensors.

- **Communication Module (MQTT):**

Handles the transmission of sensor data from IoT devices to the central system using the MQTT protocol.

- **Attack Simulation Module:**  
Implements ARP spoofing to simulate real-world Man-in-the-Middle (MITM) attacks and generate manipulated data.
- **Data Processing Module:**  
Preprocesses incoming data, including cleaning, formatting, and organizing it into sequences for analysis.
- **Feature Extraction Module:**  
Extracts important statistical and temporal features from sensor data to identify patterns.
- **Rule-Based Detection Module:**  
Detects basic anomalies such as spikes, drift, noise, and sudden drops using predefined rules.
- **Machine Learning Module:**  
Uses a Random Forest model to classify data into normal and different attack categories.
- **Replay Detection Module:**  
Identifies repeated or replayed data patterns by comparing current sequences with historical data.
- **Network Monitoring Module (Snort IDS):**  
Monitors network traffic and detects suspicious activities such as scanning and unauthorized access.
- **Decision Module (Hybrid System):**  
Combines outputs from all detection layers to generate a final decision with improved accuracy.
- **Visualization Module:**  
Displays real-time graphs and outputs for monitoring sensor data and detected anomalies.

## 3.4 Design Diagrams

### 3.4.1 Use Case Diagram

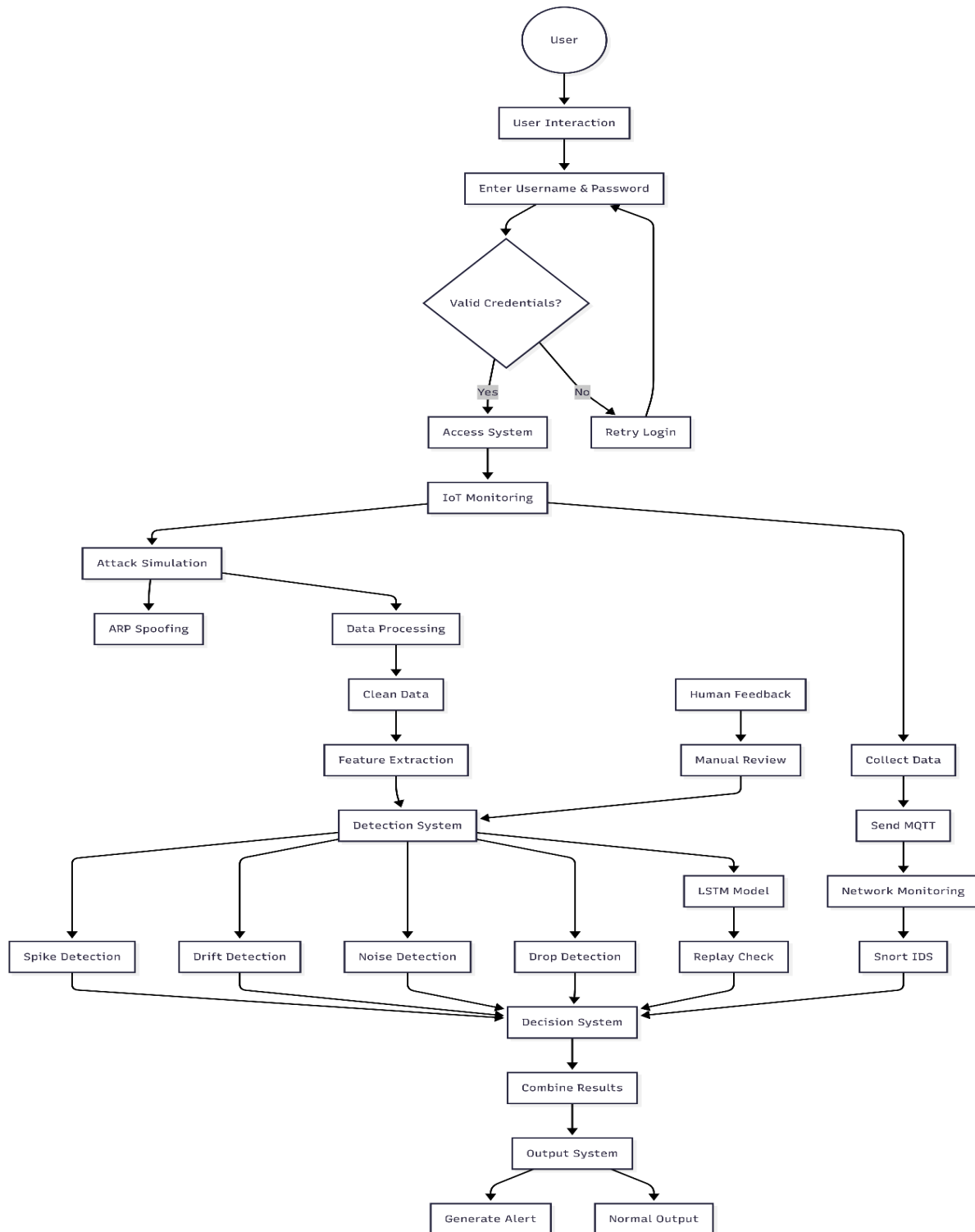


Figure 3.1

3.4.2 Activity Diagram

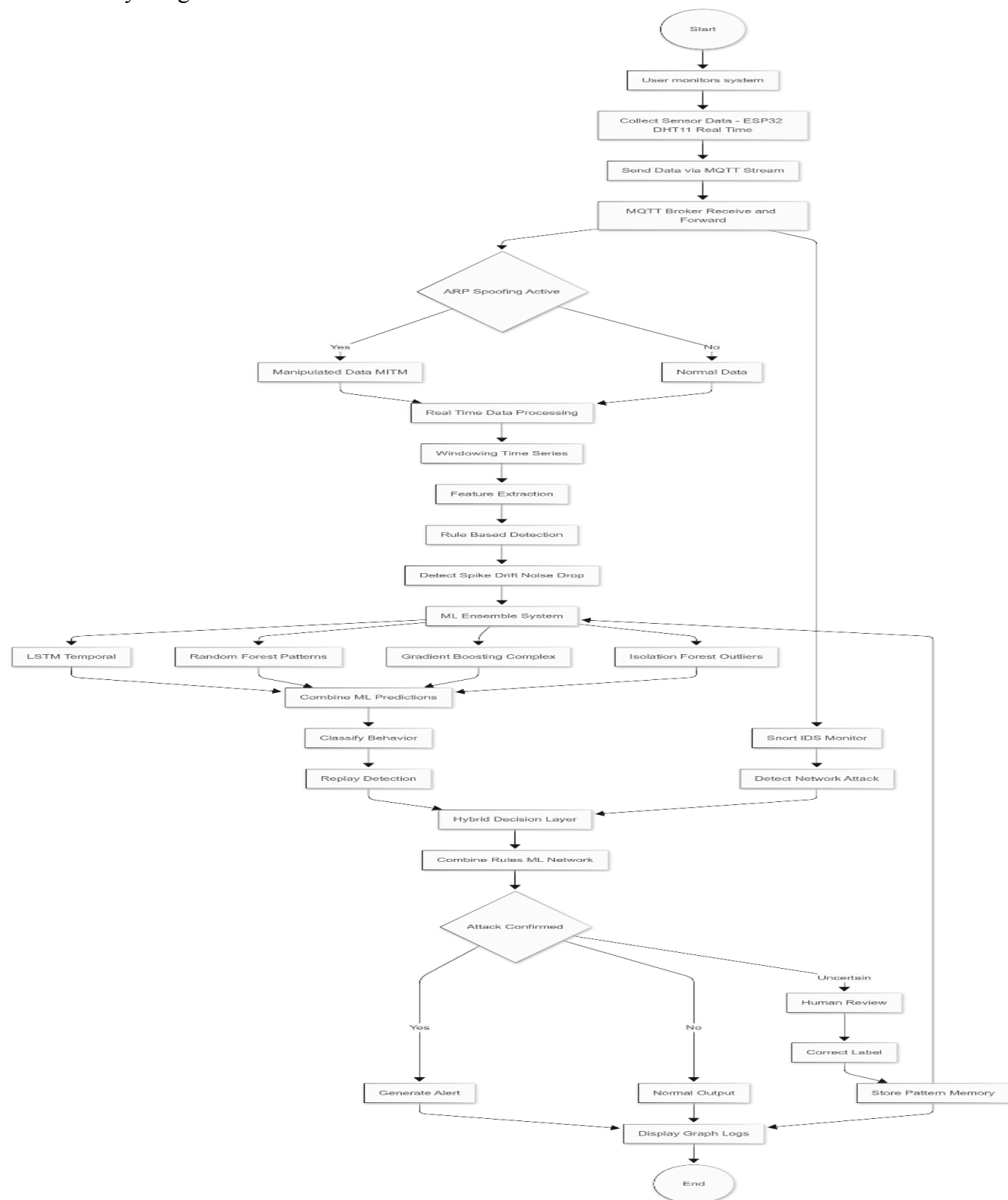


Figure 3.1

# Chapter 4

## Technology Stack and System Specifications

### 4.1 Overview

The proposed system is built using a combination of IoT hardware, communication protocols, and intelligent software components. It integrates real-time data collection, network monitoring, and anomaly detection into a unified framework. The system is designed to be lightweight, scalable, and suitable for IoT environments.

### 4.2 Hardware Components

- **ESP32 Microcontroller:**

ESP32 is a low-cost, energy-efficient microcontroller with built-in Wi-Fi and Bluetooth capabilities, widely used in IoT systems for data collection and communication

- **DHT11 Sensors:**

Used for collecting environmental data such as temperature and humidity from different zones.

- **Network Environment (Wi-Fi):**

Provides connectivity between IoT devices and the central system for real-time communication.

### 4.3 Software Components

- **Programming Language: Python**

- **Libraries: NumPy, Pandas, Scikit-learn, Matplotlib**

- **MQTT Broker (Eclipse Mosquitto):**

Used for lightweight and efficient communication between IoT devices and the processing system.

- **Snort IDS:**

An open-source network intrusion detection system used to monitor network traffic and detect suspicious activities such as scanning and unauthorized access

- **Attack Simulation Tools: Bettercap, Wireshark**

### 4.4 System Configuration

- **Multi-sensor IoT setup (3 ESP32 + DHT11 nodes)**

- **Real-time data transmission using MQTT protocol**

- **Dataset generated through real-time attack simulation (ARP spoofing)**

- **Machine learning model: Random Forest Classifier**

- **Feature engineering using statistical and temporal features**

- **Window-based data processing for sequence analysis.**

## **4.5 Design Considerations**

### **4.5.1 Scalability**

The system is designed to support additional IoT devices and sensors without significant changes to the architecture.

### **4.5.2 Reliability**

Multiple detection layers (rule-based, ML, and network monitoring) ensure consistent and accurate results even under attack conditions.

### **4.5.3 Security**

The system addresses both network-level and data-level threats, including ARP spoofing and data manipulation attacks.

# Chapter 5

## Implementation

### 5.1 Environment Setup

The system was developed using a combination of hardware and software tools. ESP32 microcontrollers were connected with DHT11 sensors to collect environmental data. All devices were connected over a common Wi-Fi network.

The software environment included Python for data processing and machine learning, Eclipse Mosquitto as the MQTT broker for communication, and Snort IDS for network monitoring. Tools such as Bettercap and Wireshark were used for attack simulation and traffic analysis.

### 5.2 Data Collection and Processing

Real-time environmental data such as temperature and humidity was collected from multiple sensor nodes. The data was transmitted to the MQTT broker and then received by a Python-based subscriber system.

The collected data was cleaned and structured into a dataset. It was further processed into fixed-size windows to capture temporal behavior. Feature extraction techniques were applied to generate meaningful attributes such as mean, standard deviation, and rate of change.

### 5.3 Attack Simulation

A real-world attack scenario was implemented using ARP spoofing. The attacker system intercepted communication between ESP32 devices and the MQTT broker.

Using this approach, sensor data was manipulated in real time to simulate different attack patterns such as spikes, drift, and repeated data. This helped in generating realistic datasets for training and testing the detection system.

### 5.4 Model Development

A Random Forest classifier was developed for anomaly detection. The model was trained using the generated dataset containing both normal and attack data.

The system used temporal window-based learning instead of individual data points. The model classified data into different categories such as normal behavior, injection, drift, noise, drop, and replay attacks.

Feature scaling and preprocessing were applied to improve model performance and accuracy.

### 5.5 System Integration

All components of the system were integrated into a unified framework. Sensor data collection, attack simulation, detection mechanisms, and network monitoring were combined into a hybrid detection pipeline.

The system processed incoming data in real time, applied rule-based and machine learning detection, and correlated results with network-level alerts from Snort ID.

## **Chapter 6**

### **Output Screens and Results**

Figure 6.1:

# Chapter 7

## Limitations and Future Enhancements

### 7.1 Limitations

The system currently operates in a controlled environment, and real-world deployment may introduce additional challenges such as network instability and device heterogeneity.

- The machine learning model performance depends on the quality and diversity of the dataset. Limited variations in attack scenarios may affect generalization.
- Real-time processing may introduce slight delays when handling large volumes of data or multiple sensor nodes simultaneously.
- The system relies on predefined rules for certain detections, which may not cover all possible attack patterns.
- Snort IDS is used in detection mode only and does not actively block malicious traffic.

### 7.2 Future Enhancements

#### 7.2.1 Technical Improvements

- Integration of deep learning models (such as LSTM or CNN) to improve detection of complex temporal patterns.
- Implementation of automated response mechanisms (IPS) to block malicious activities in real time.
- Expansion of the dataset with more real-world attack scenarios to improve model robustness.
- Optimization of the system for large-scale deployments with multiple IoT devices.
- Development of a dashboard-based interface for advanced monitoring and control.

#### 7.2.2 Applications

- **Smart Buildings:** Enhancing reliability and security of automated HVAC and environmental systems.
- **Industrial IoT (IIoT):** Monitoring and protecting industrial sensor networks from cyber threats.
- **Healthcare Systems:** Ensuring secure monitoring of medical IoT devices and patient data.
- **Smart Cities:** Securing large-scale IoT infrastructures used in traffic, energy, and surveillance systems.
- **Research and Cybersecurity Testing:** Providing a platform for studying IoT attack scenarios and developing advanced detection techniques.

# Chapter 8

## Conclusion

This project presents a hybrid intrusion detection system designed for IoT-based smart building environments. The system addresses a critical issue in modern automated systems, where decisions are made based on sensor data that may be incorrect or manipulated.

The proposed solution integrates multiple components including IoT-based data collection, real-time communication using MQTT, network monitoring through Snort IDS, and intelligent anomaly detection using rule-based techniques and machine learning. A real-world attack scenario using ARP spoofing was implemented to generate realistic datasets and evaluate system performance.

By combining data-level analysis with network-level monitoring, the system provides a more comprehensive approach to intrusion detection compared to traditional methods. The use of temporal analysis and a hybrid decision-making mechanism improves detection accuracy while reducing false positives.

The results demonstrate that the system is capable of identifying various types of anomalies and attacks in real time, making it suitable for practical deployment in IoT environments. Overall, the project highlights the importance of validating both data security and data correctness in cyber-physical systems.

The proposed system provides a scalable and effective solution for improving the reliability and security of smart building systems and can be extended further to support advanced detection and prevention mechanisms in the future.

# Appendix A

## Keywords

| Keyword                          | Definition                                                                                                         |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| IoT (Internet of Things)         | A network of interconnected devices that collect and exchange data in real time.                                   |
| Intrusion Detection System (IDS) | A system that monitors network or system activities to detect malicious behavior or security threats.              |
| MQTT Protocol                    | A lightweight messaging protocol used for communication between IoT devices and servers.                           |
| ESP32                            | A low-cost microcontroller with built-in Wi-Fi used for IoT-based data collection and communication.               |
| DHT11 Sensor                     | A sensor used to measure temperature and humidity in IoT environments.                                             |
| ARP Spoofing                     | A type of cyber-attack where an attacker intercepts communication by manipulating network address mappings.        |
| Machine Learning                 | A method that enables systems to learn patterns from data and make predictions or classifications.                 |
| Random Forest                    | A machine learning algorithm that uses multiple decision trees for accurate classification.                        |
| Anomaly Detection                | The process of identifying unusual patterns that do not match expected system behavior.                            |
| Rule-Based Detection             | A method that uses predefined rules to identify specific types of anomalies or attacks.                            |
| Replay Attack                    | An attack where previously captured data is present to manipulate system behavior.                                 |
| Snort IDS                        | An open-source network intrusion detection system used to monitor and analyze network traffic.                     |
| Feature Engineering              | The process of extracting meaningful features from raw data for analysis and model training.                       |
| Temporal Analysis                | The analysis of data over time to detect patterns and changes in behavior.                                         |
| Hybrid Detection System          | A system that combines multiple detection techniques such as rule-based, machine learning, and network monitoring. |

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